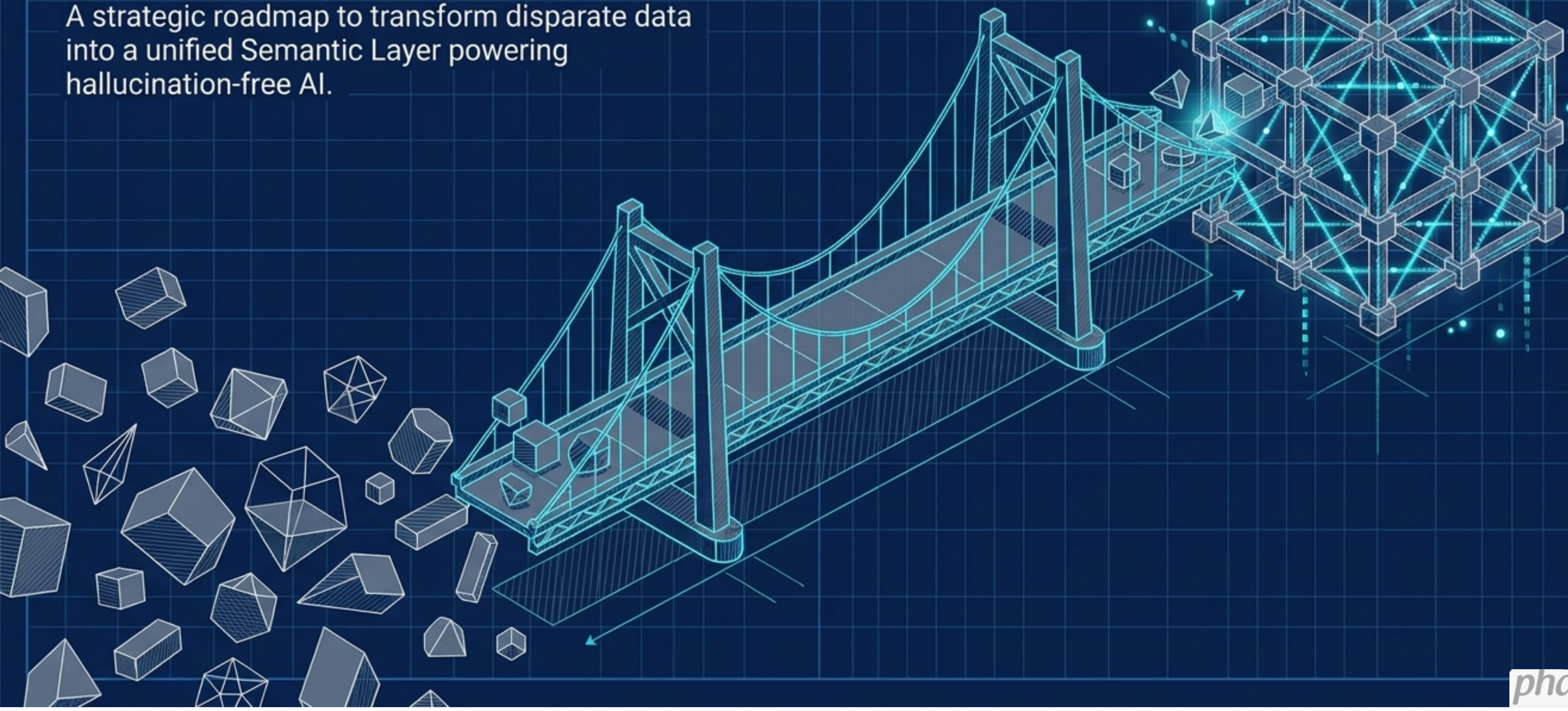


# Building the Architectural Bridge for Pharma R&D

A strategic roadmap to transform disparate data  
into a unified Semantic Layer powering  
hallucination-free AI.





# The Competitive Mandate: Moving Beyond the Data Swamp



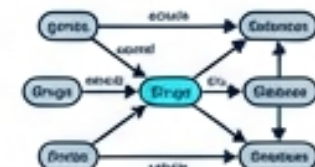
## The Old Paradigm (Data-Driven)

- **Infrastructure:** Dirty Data Swamps — massive, siloed collections of raw data points.
- **AI State:** Probabilistic LLM guesses based on statistical co-occurrence. High hallucination risk.
- **Operations:** Integration debt accumulates; months of manual curation required for single targets.
- **Identity:** Strings of text trapped in disparate legacy systems.



## The ARCH Paradigm (Knowledge-Driven)

- **Infrastructure:** Pharma Repository of Knowledge (ARK) — a single source of verified semantic truth.
- **AI State:** Grounded RAG 2.0 generating deterministic, verified Cypher database queries.
- **Operations:** Resolves integration debt via recursive identity preservation; insights delivered in minutes.
- **Identity:** Universal biological things with causal relationships explicitly mapped.





# The Pharma R&D Convergence Hub (ARCH)

## The Goal

Act as the architectural bridge between vast, siloed R&D data and AI-driven insights across Oncology, Immunology, and Neuroscience.



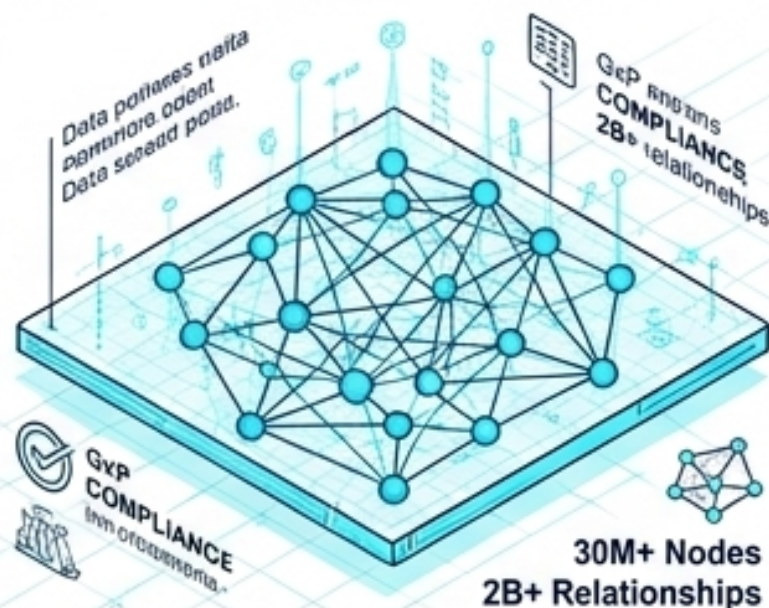
## The Method

Apply unyielding Model-Based Systems Engineering (MBSE) principles to guarantee zero data loss.



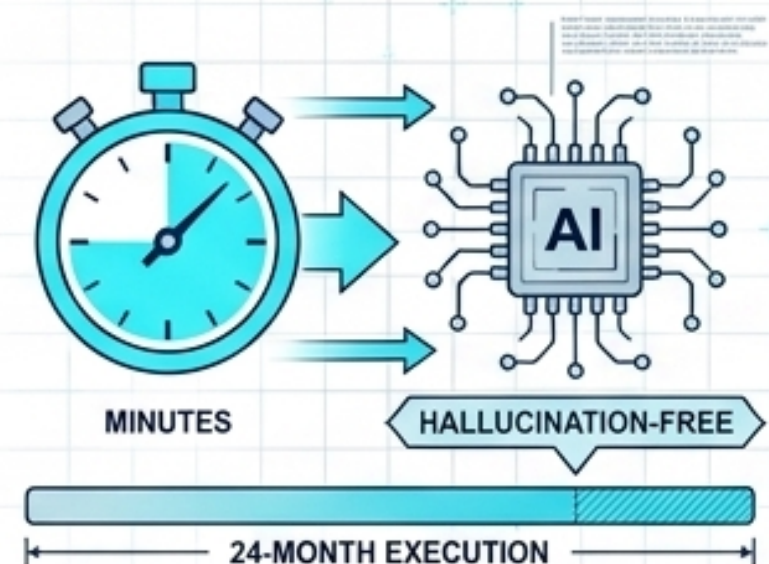
## The Output

Transform raw data points into a robust, GxP-compliant Semantic Layer holding 30M+ nodes and 2B+ relationships.



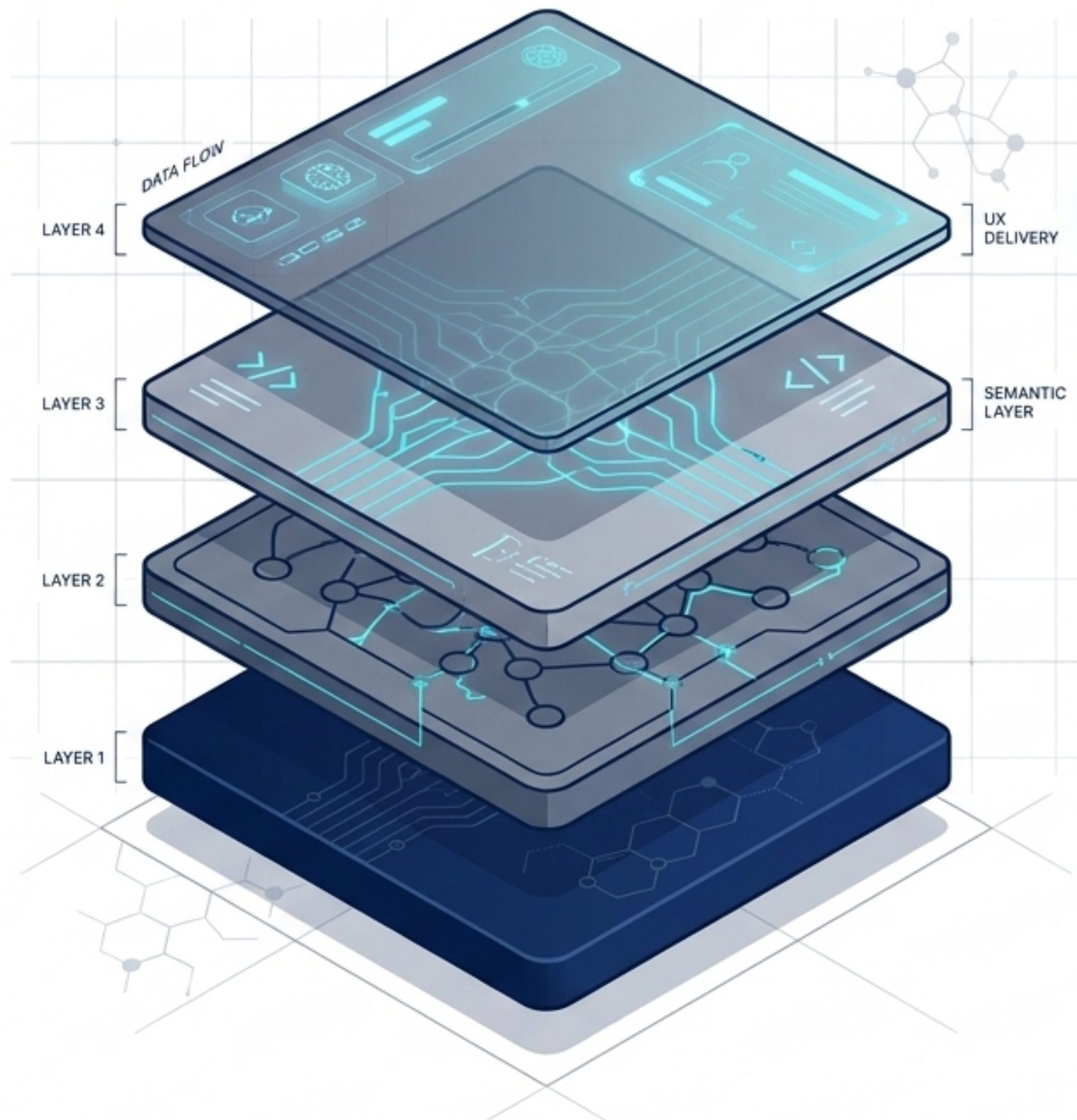
## The Impact

Compress discovery timelines from months to minutes and power hallucination-free AI over a precise 24-month execution timeline.





# The Architecture of Truth: A Four-Layer Blueprint



## Layer 4: User Interface & Application Delivery (Mendix / Unity)

Component-based microapps delivering tailored experiences to specific personas at the exact point of consumption.

## Layer 3: AI & Prompt Engineering (AWS Bedrock)

The Semantic Gateway employing Federated RAG 2.0 to guarantee grounded, hallucination-free outputs.

## Layer 2: Semantic Harmonization (Neo4j / Tellic)

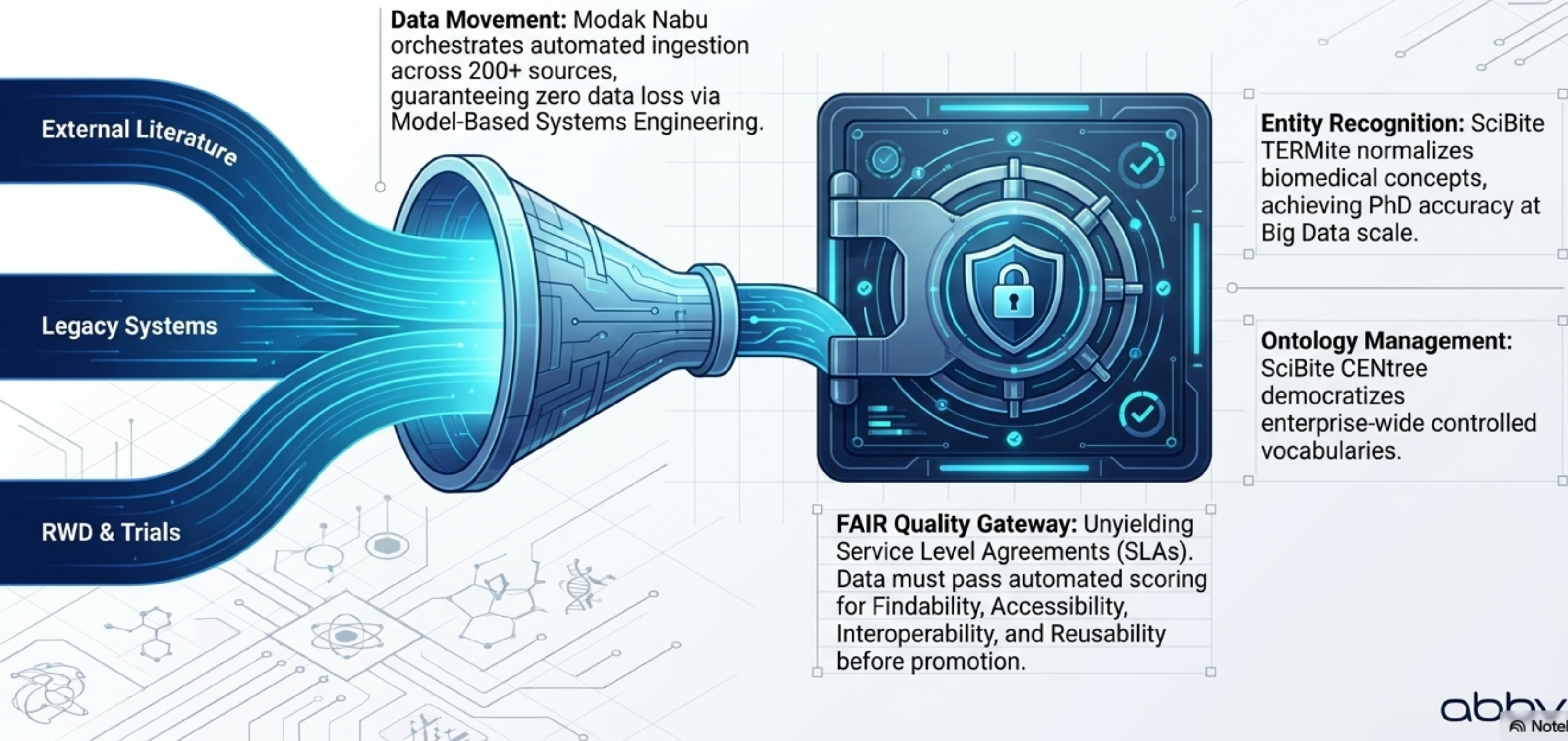
Translates disparate strings into unified biological things within a labeled property graph.

## Layer 1: Data Foundation & Ingestion (Modak Nabu / SciBite)

MBSE-governed ingestion centralizing 124TB of data from over 200 distinct sources and legacy systems.



# Layer 1: MBSE-Governed Data Foundation





## Layer 2: Semantic Harmonization and the ARK Graph

Tumor  
Neoplasm  
Tumor  
Neoplasm  
C0027651



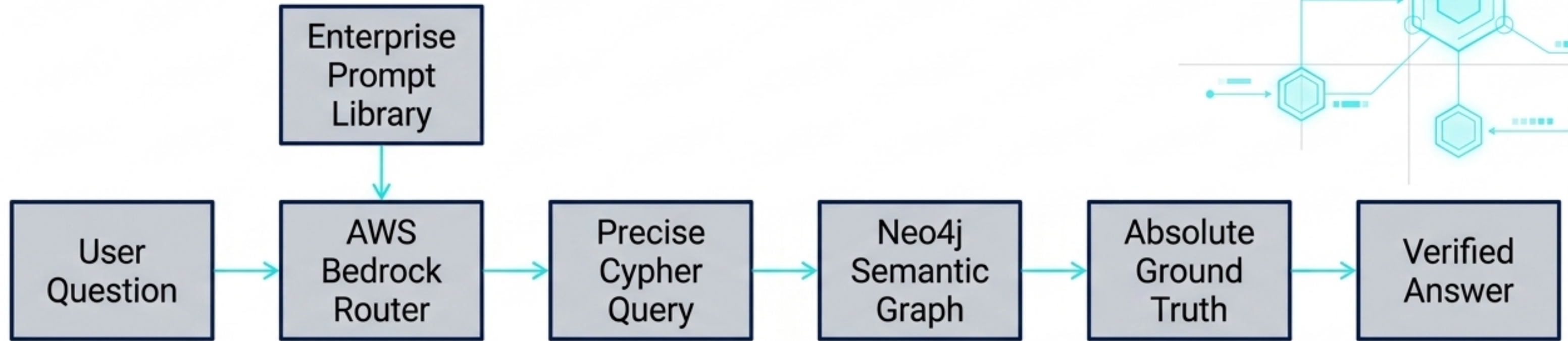
Neo4j Semantic Engine



- |                                                                                                                                                                                                                                    |                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• <b>Relationship Extraction:</b> Telic extracts 1B+ biomedical relationships from global literature using distributional semantics, moving beyond statistical co-occurrence.</li></ul>      | <ul style="list-style-type: none"><li>• <b>The Fidelity Gate:</b> A strict 0.9 confidence threshold is applied to all NLP-derived triples to prevent the backfire of erroneous data.</li></ul> |
| <ul style="list-style-type: none"><li>• <b>The Engine:</b> A Neo4j graph database serves as the core Pharma Repository of Knowledge (ARK), managing 30M+ nodes and 80+ relationship types (e.g., inhibits, upregulates).</li></ul> | <ul style="list-style-type: none"><li>• <b>Entity Resolution:</b> The Reveal-Reframe-Resist model identifies NLP failures, mapping concepts precisely to ontologies like UniProt.</li></ul>    |



# Layer 3: The Semantic Gateway & RAG 2.0



## Enterprise Router:

AWS Bedrock routes queries to specialized language models based on user persona, permissions, and therapeutic context.

## Intent-Driven Prompting:

An Enterprise Prompt Library wraps requests in ontological metadata before reaching the LLM.

## Federated RAG 2.0:

AI interacts directly with the Neo4j graph, generating deterministic Cypher queries instead of probabilistic guesses.

## The Result:

Answers are strictly grounded in factual, traceable data, completely mitigating the risk of hallucination.



# Layer 4: Component-Based Application Delivery

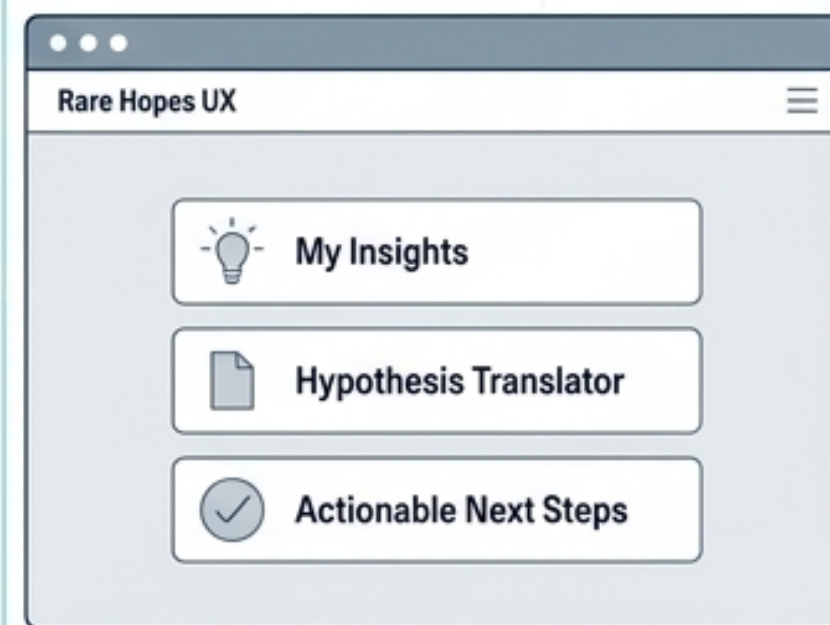
Built on Mendix low-code architecture and the Unity design system to deliver 18 distinct views of a biological target in minutes, not months.



**For Scientists:**  
**Ask ARCH & Target/Safety Dossiers**  
A natural language interface eliminating the need to understand complex graph database structures.



**For Practitioners:**  
**Pharma Science Oncology Platform**  
A modernized portal delivering tailored, graph-grounded research updates directly to Healthcare Professionals.



**For Patients:**  
**Rare Hopes UX**  
Translates complex graph hypotheses into accessible, actionable insights for rare disease shared decision-making.

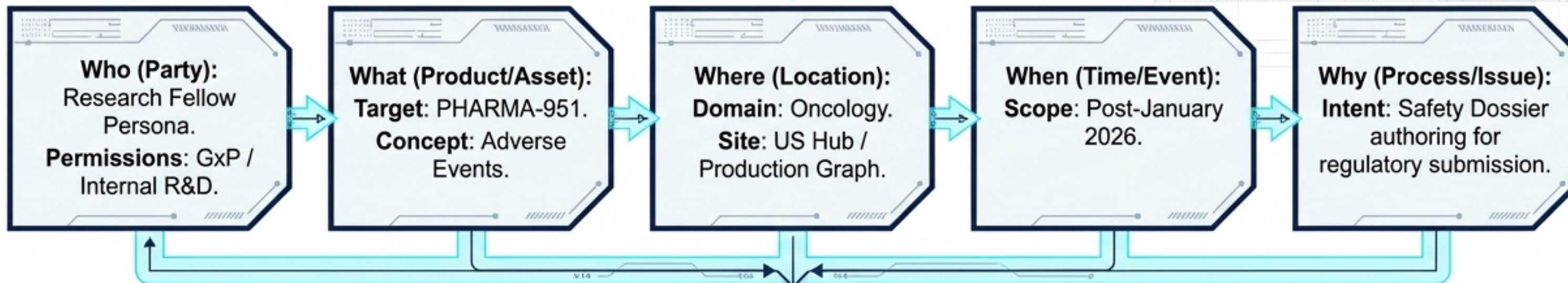


# Operationalizing Relevancy: The Context Engine

## 1. Raw Input:

Raw Input: "What is the latest on drug safety for our new oncology target?"

## 2. The 5 Pillars of Ontological Extraction:

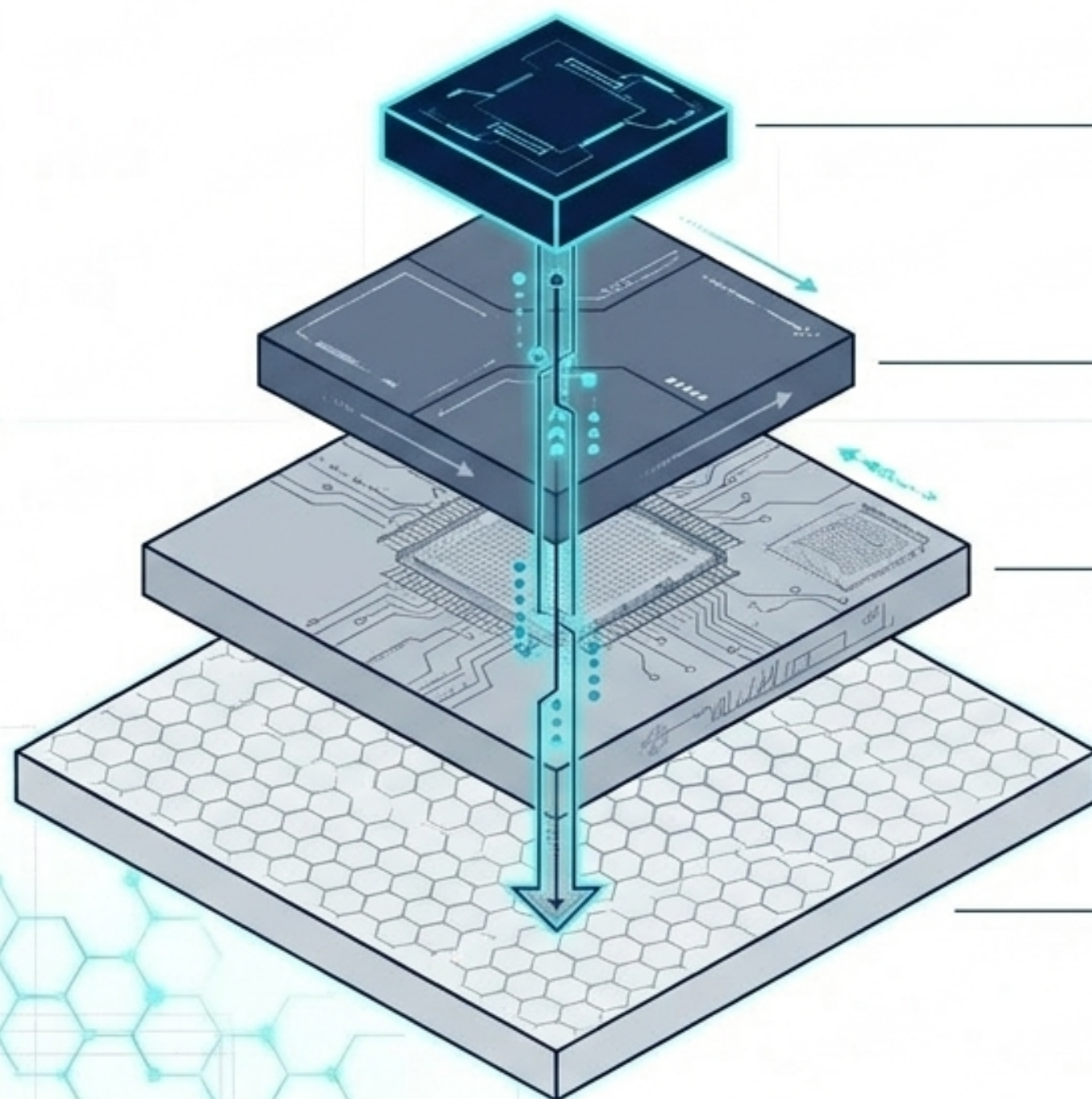


## 3. Verified Cypher Generation:

```
MATCH (p:Product {name: 'PHARMA-951'})-[:HAS_TARGET]->(t:BiologicalTarget)
MATCH (t)-[r:ASSOCIATED_WITH_ADVERSE_EVENT]-(ae:Issue)
WHERE r.timestamp > '2026-01-01' AND r.source = 'Internal_Clinical_Trial'
RETURN p.name, t.name, ae.description, r.severity, r.timestamp
```



# Federated Semantic Governance: The 4-Level Topology



**Level 1: The Constitutional Layer (Altair Graph Studio)**  
The Core 9 Unifying Ontologies (Process, Service, Asset, Product, Party, Location, Time, Event, Issue). Enterprise-wide, intentionally stable semantic ceiling.

**Level 2: Business Domain Ontologies (Altair Graph Studio)**  
Organizational layer specifying the Core for specific business units (e.g., Oncology R&D).

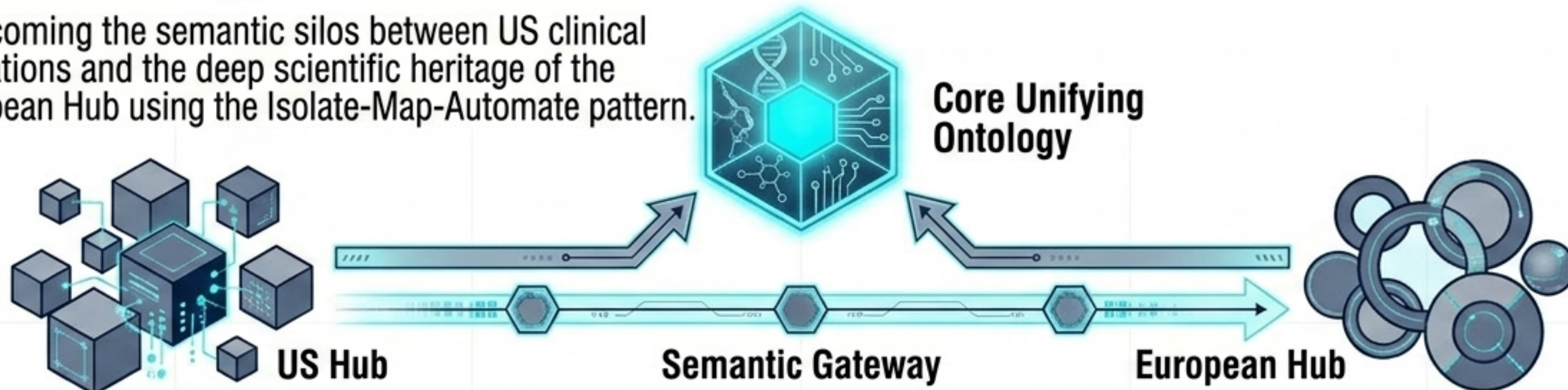
**Level 3: The Knowledge Warehouse (SciBite CENTree)**  
Dense, multi-layered scientific hierarchies (e.g., Kinase Targets, Immune Pathways). Local curation by domain scientists with global alignment.

**Level 4: The Digital Mirror (Modak Nabu / Neo4j)**  
Bottom-up source schemas and the physical graph topology representing operational reality. Connected via AI-driven Mapping Agents.



# Cross-Site Semantic Bridging: Resolving Integration Debt

Overcoming the semantic silos between US clinical operations and the deep scientific heritage of the European Hub using the Isolate-Map-Automate pattern.



## 1. Isolate

European scientists maintain specific proprietary terminology (e.g., Melt Extrusion formulation parameters) in standalone SciBite CENTree Application Ontologies. No cross-domain clutter.

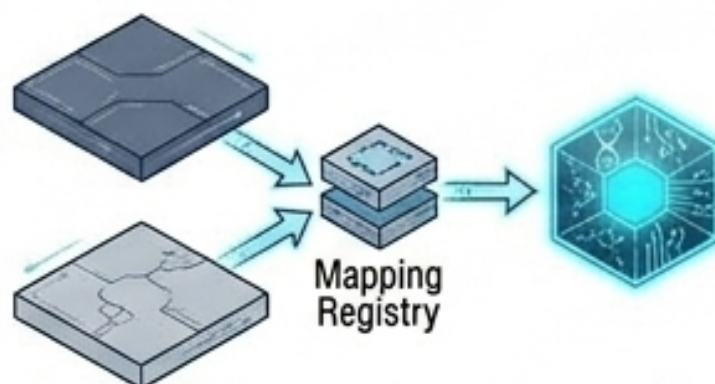


Specific Terminology



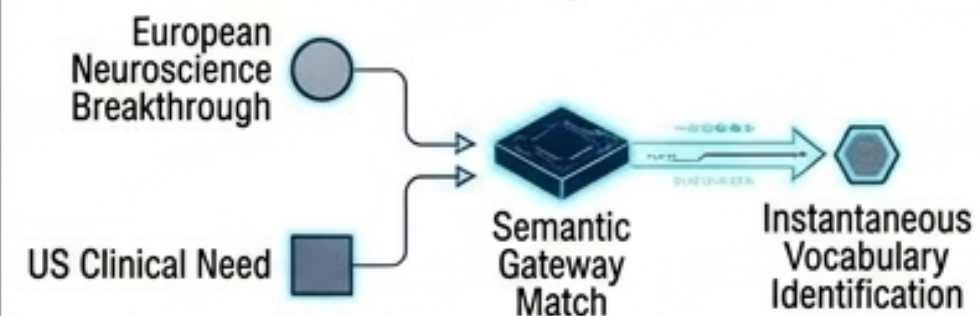
## 2. Map

Local terms are linked to the stable Core Unifying Ontology via the Mapping Registry, creating a Universal Transiator.



## 3. Automate

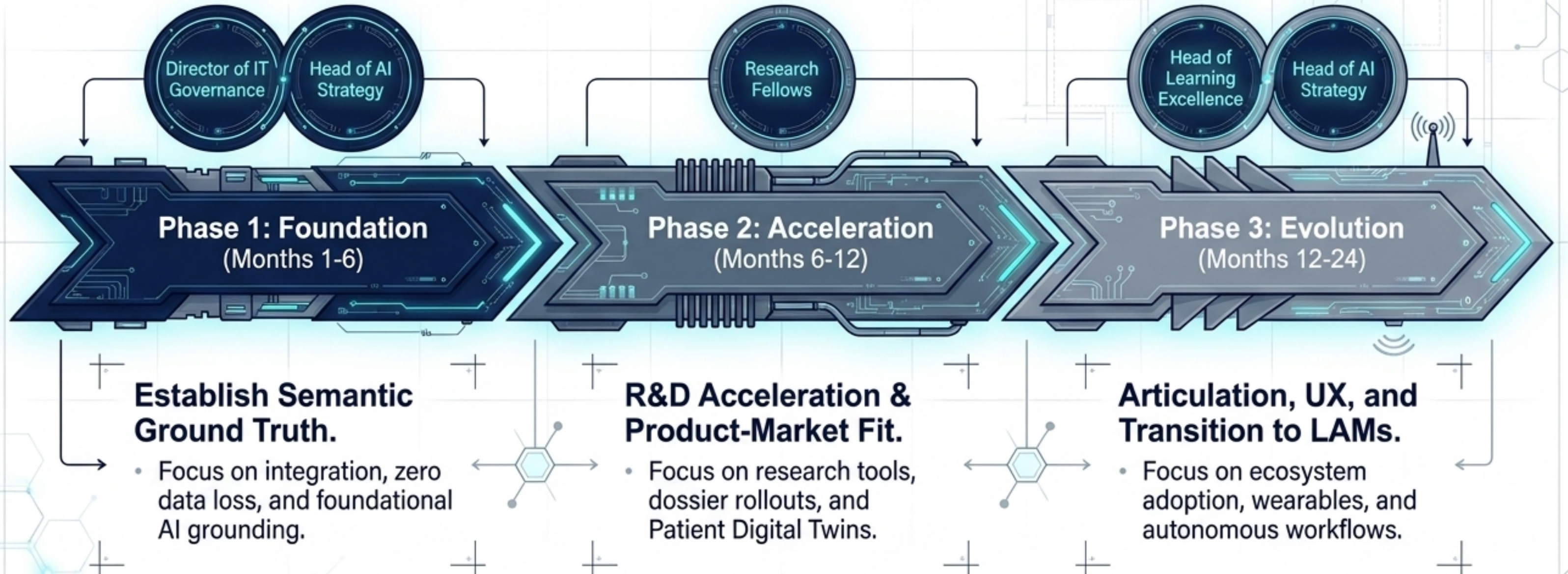
When a European neuroscience breakthrough matches a US clinical need, the Semantic Gateway instantly identifies the match across vocabularies without manual data hunting.





# The 24-Month Execution Roadmap

A 'Quick but not Cheap' mandate structured to deliver incremental business value while advancing toward autonomous agentic intelligence.





# Phase 1: Foundation (Months 1–6)

Solidifying Integration and  
Establishing Ground Truth

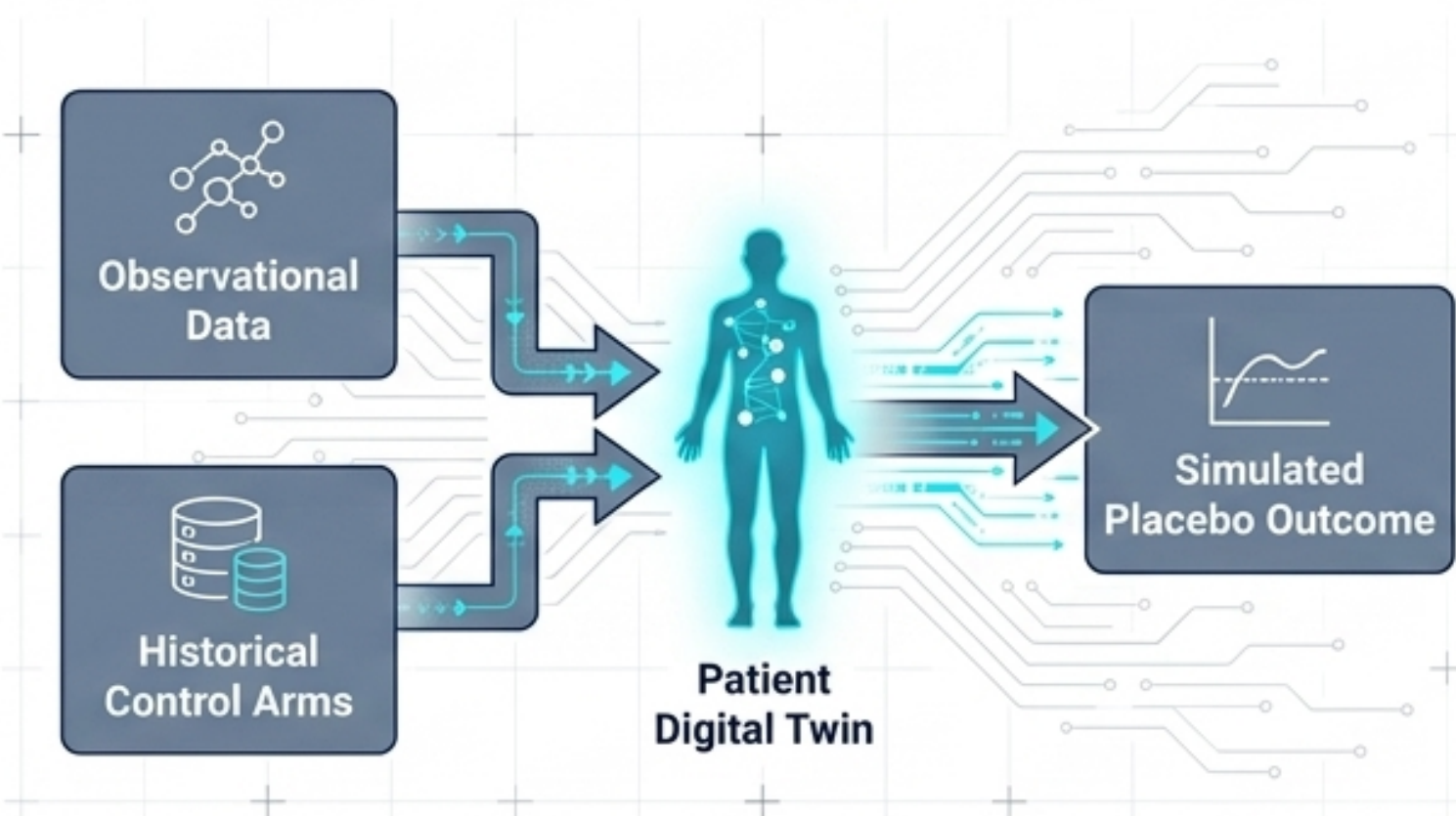


- **Unified Data Lake:** Connecting 200+ internal and external sources.
- **MBSE Alignment:** Applying Model-Based Systems Engineering pipelines to guarantee structurally sound integration and zero data loss.
- **Outcome:** Achieving PhD accuracy at Big Data scale.
- **FAIR Quality Gateway:** Enforcing Findability, Accessibility, Interoperability, and Reusability to achieve 100% compliance.
- **Building the Guardrails:** Populating the ARK property graph (30M+ nodes) and implementing initial RAG frameworks connecting Neo4j to LLMs.



# Phase 2: Acceleration (Months 6–12)

Driving Scientific Product-Market Fit



## Deploying Patient Digital Twins

- Integrating real-world observational data and historical control arms into patient-specific computational models.
- Accurately simulating placebo outcomes to optimize clinical trial sample sizes and dramatically accelerate time-to-market.



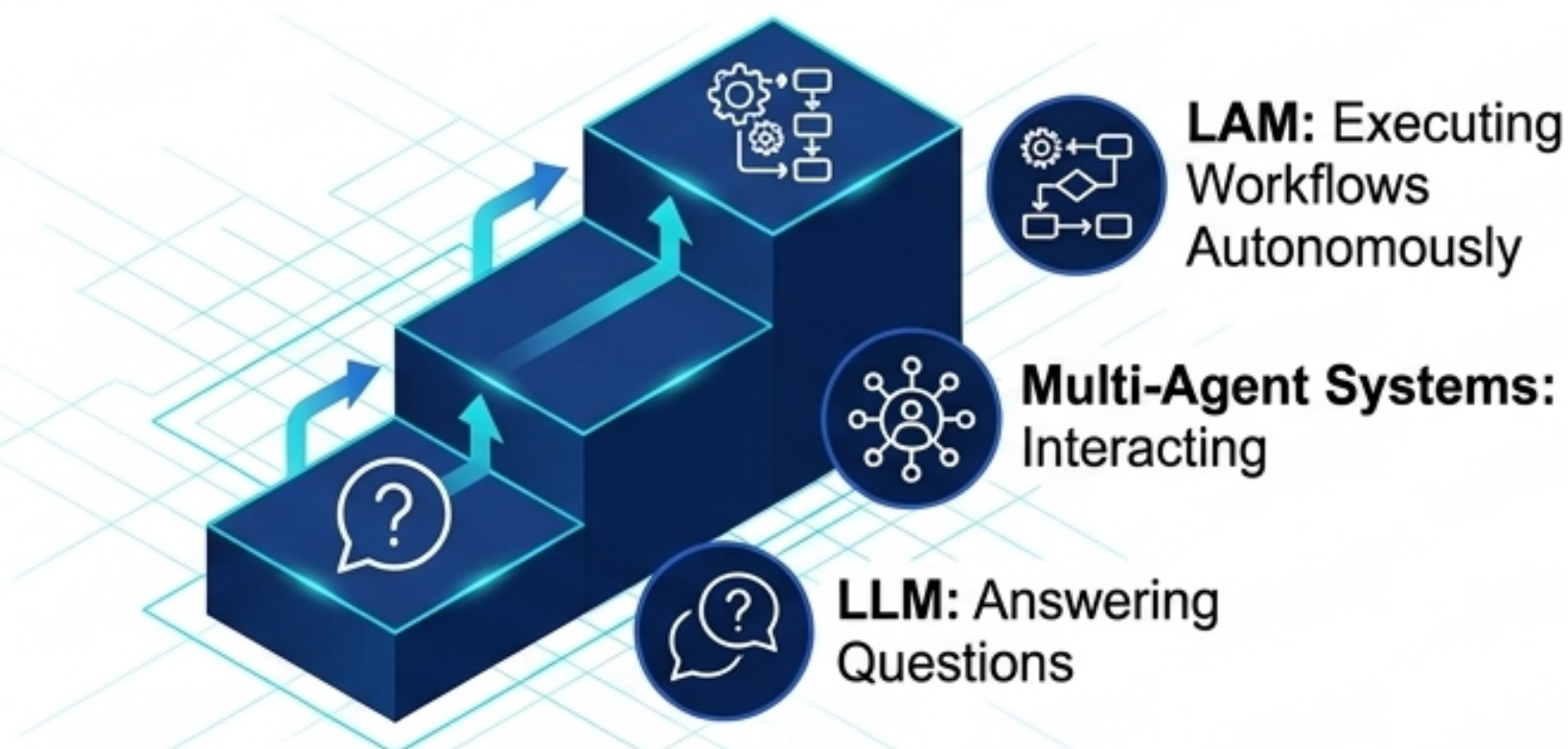
## The Research Accelerator

- Transitioning the ontology into high-value products via Mendix microapps.
- Dossier Rollouts:** Delivering 18 distinct views, reducing time-to-insight from months to minutes.
- Ask ARCH:** Releasing conversational graph querying to the broader scientific community.



# Phase 3: Evolution (Months 12–24)

## Autonomous Workflows and Real-World Endpoints



### From Language Models to Action Models (LAMs)

- Evolving beyond Q&A to Large Action Models that autonomously execute multi-step R&D workflows (hypothesis generation to experimental design).
- The Gaia Platform: A recursive learning environment where the ontology actively executes and enriches the graph based on experimental feedback loops.

### Digital Health Technology (DHT) Scaling

- Integrating continuous physiological data streams from wearable devices directly into the FAIR-compliant ARCH foundation.
- Maturing Patient Digital Twins into living computational models for real-world evidence generation.



# Strategic Risk Mitigation for the Enterprise

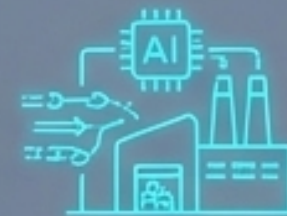
## Debt & Capital Risk

Optimizes clinical trial designs via Patient Digital Twins, reducing redundant experiments and late-stage attrition before capital is committed.



## Execution Risk

Acts as the validated pipeline feeder for new hundreds-of-millions-dollar AI-powered manufacturing facilities, ensuring capacity is utilized for high-fidelity targets.



## Margin Stability

Compresses discovery timelines and generates real-world evidence via wearable DHT endpoints to support value-based pricing strategies.



## Regulatory Pressure

Context Engine auditability captures the Who, What, Where, When, and Why of every AI-generated insight, delivering fundamentally GxP-ready documentation.

| AUDIT TRAIL      |            |               |
|------------------|------------|---------------|
| TIME             | USER/ID    | DESCRIPTION   |
| 2023-10-27 10:00 | DR. J. DOE | VIEWED REPORT |
| 2023-10-27 10:05 | DR. J. DOE | EDITED REPORT |
| 2023-10-27 10:10 | DR. J. DOE | VIEWED REPORT |
| 2023-10-27 10:15 | DR. J. DOE | EDITED REPORT |
| 2023-10-27 10:20 | DR. J. DOE | VIEWED REPORT |
| 2023-10-27 10:25 | DR. J. DOE | EDITED REPORT |



# A Fully Realized Ecosystem Powering the Future of Drug Discovery



## The Result of ARCH

A seamlessly integrated, zero-data-loss environment governed by MBSE, resolving historical integration debt.

## Uncompromised AI

A platform where every AI interaction is grounded in absolute semantic truth, free of hallucinations and traceable to the source row.

## Accelerated Impact

Translating complex biological relationships into accelerated clinical trials, autonomous workflows, and ultimately, faster delivery of life-changing treatments to patients.